

**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING**

Khwopa College Of Engineering
Libali, Bhaktapur
Department of Computer Engineering



**A PROPOSAL ON
FACIAL RECOGNITION SYSTEM ALONG WITH
REAL TIME TRACKING**

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF COMPUTER ENGINEERING

Submitted by

Gobinda Bista

KCE077BCT015

Hari Krishna Khuju

KCE077BCT016

Subash Bhandari

KCE077BCT037

Suresh Shrestha

KCE077BCT040

Under the Supervision of

Er.Sushil Dyopala

Department Of Computer Engineering

Khwopa College Of Engineering
Libali, Bhaktapur

Abstract

Facial recognition technology has witnessed remarkable advancements in recent years, emerging as a pivotal tool across diverse industries for identity verification and access control. The proposed system aims to enhance security measures, streamline access control, and provide valuable insights into user behaviors across various applications. This proposal introduces a comprehensive system that integrates multi-target facial recognition with real-time tracking capabilities. Leveraging cutting-edge computer vision algorithms, deep learning-based face recognition algorithms and real-time tracking software, this system offers a versatile solution applicable in settings ranging from secure facilities to retail environments. The system architecture incorporates facial recognition algorithms, a database of facial templates, high-quality camera systems, and real-time tracking software. By providing instant identification and continuous tracking of recognized individuals, the system offers a robust security mechanism and the ability to monitor user activities in real-time.

Keywords: Face detection, Face recognition, Tracking multiple face

Contents

Abstract	2
Contents	2
List of Figures	4
List of Symbols and Abbreviation	6
1 Introduction	1
1.1 Background Introduction	1
1.2 Problem Statement	2
1.3 Objective	2
2 Literature Review	3
3 FEASIBILITY STUDY	5
3.0.1 Technical Feasibility	5
3.0.1.1 Requirements Analysis	5
3.0.1.2 System Compatibility	5
3.0.1.3 Facial Recognition Technology	5
3.0.2 Economic Feasibility	5
3.0.2.1 Cost-Benefit Analysis	5
3.0.2.2 Return on Investment (ROI)	5
3.0.3 Operational Feasibility	6
3.0.3.1 User Acceptance	6
3.0.3.2 Training Requirements	6
3.0.3.3 Integration with Existing Processes	6
4 Methodology	7
4.1 Software Development Methodology	7
4.1.1 Agile Methodology	7
4.2 ALGORITHMS AND MODELS	8
4.2.1 Haar Cascades	8
4.2.2 Local Binary Patterns	9
4.2.3 Eigenfaces	9
4.2.4 Local Feature Analysis	9
4.2.5 Convolutional Neural Networks	9
4.3 Proposed System	9
4.3.1 Dataset Creation and Image Preprocessing	10
4.3.2 Face Detection	10
4.3.3 Face Recognition and Calculation	10
4.3.4 DataBase	11

4.4	System (or Project) Design and Architecture	11
4.5	USE CASE DIAGRAM	11
4.6	System Block	12
4.7	Sequence Diagram	12
4.8	Class Diagram	13
5	Implementation Plan	14
5.1	Schedule(Gantt-Chart)	14
5.2	SOFTWARE REQUIREMENT	14
5.2.1	Python	14
5.2.2	OpenCV (Open Source Computer Vision Library)	15
5.2.3	MySQL	15
5.2.4	Tensor Flow	15
5.3	FUNCTIONAL REQUIREMENT	15
5.3.1	User Registration	15
5.3.2	Face Detection	15
5.3.3	Face Recognition	15
5.3.4	Enrollment and Database Management	16
5.3.5	Real-time Monitoring	16
5.4	NON-FUNCTIONAL REQUIREMENT	16
5.4.1	Performance	16
5.4.1.1	Response Time	16
5.4.1.2	Throughput	16
5.4.2	Scalabiity	16
5.4.3	Reliability	16
5.4.4	Availability	16
5.4.5	Security	16
5.4.5.1	Data Security	16
5.4.5.2	Access Control	17
6	Expected Outcomes	18
6.1	Conclusion	18

List of Figures

4.1	System architecture of face recognition system	9
4.2	System Use Case Diagram	11
4.3	System Block Diagram	12
4.4	Sequence Diagram	12
4.5	Class Diagram	13
5.1	Gantt Chart	14

List of Symbols and Abbreviation

CNN	Convolution Neural Network
DNN	Deep Neural Network
SVM	Support Vector Machine
MATLAB	Matrix Laboratory
SMPC	Shared Memory Parallel Computer
LDA	Linear Discriminant Analysis
RFID	Radio Frequency Identification
GUI	Graphical User Interface
PCA	Prompt Corrective Action
MLP	Multi-Layered Perception
HD	High Definition
HOG	Head Of Generation
LBP	Local Binary Patterns
ROI	Return on Investment
ROIs	Region Of Interface
SQL	Structured Query Language
LFA	Local Features Analysis
OpenCV	Open Source Computer Vision Library
RDBMS	Reational Database Managaement System
PHP	Hypertext Preprocssor

Chapter 1

Introduction

1.1 Background Introduction

In an era marked by rapid technological advancement, the fusion of facial recognition and real-time tracking stands at the forefront of transformative innovations. Facial recognition technology, with its roots in computer vision and artificial intelligence, has evolved into a powerful means of biometric identification. Concurrently, real-time tracking adds a dynamic dimension to this capability, enabling continuous monitoring and analysis of individuals' movements. This fusion of facial recognition and real-time tracking holds immense potential across a spectrum of industries, reshaping security protocols, refining access control mechanisms, and revolutionizing user engagement experiences.

Attendance tracking is a critical aspect of organizational management. Automated Attendance tracking has been applied to many industries as it offers more efficient and secure method for recording employee attendance. Traditional methods often are tedious, prone to errors and fails to handle large data. Hence, in recent years, there have been much interests in Automated Attendance tracking system. Therefore, many institutes started deploying many other techniques for recording attendance like use of Radio Frequency Identification (RFID) [3], iris recognition [4], fingerprint recognition, and so on. However, these systems are queue based which might consume more time and are intrusive in nature.

Among them Facial Recognition has set an important biometric features. Face recognition technology is better than other biometric based recognition techniques like finger-print, palm-print, iris because of its non-contact process. Recognition techniques using face recognition can also recognize a person from a distance, without any contact or interaction with person. This proposal aims to implement a Facial Recognition Attendance System using OpenCV, a widely-used computer vision library, to enhance accuracy and efficiency in attendance management. The purpose of this system is to streamline and enhance our attendance tracking process.

1.2 Problem Statement

The current manual attendance tracking system is prone to errors, time-consuming, and lacks efficiency. Employees may forget to sign in, leading to inaccurate records. The manual process of collecting attendance data for all your employees will take time. You might also end up spending a lot of time maintaining scorecards, rectifying errors in time entry, and so on. Currently most of the facial recognition techniques is able to work fine only if the number of people in one frame is very few and under controlled illumination, proper position of faces and clear images. The proposed Facial Recognition Attendance System addresses these issues by providing

Accuracy: Facial recognition minimizes the risk of errors associated with manual data entry.

Efficiency: Real-time processing ensures quick and efficient attendance tracking.

Security: Facial recognition adds a layer of security, preventing unauthorized access and ensuring the integrity of attendance records

1.3 Objective

Chapter 2

Literature Review

In this paper, authors have used deep learning methods for the fiducial points extraction and embedding. The best results were obtained using a customized architecture of residual network with 18 layers, drastically reducing the number of parameters from 19M to 700k. Support Vector Machine (SVM) is used for classification task since it is fast for both training and inference. The system achieved an error rate of 0.12103 for facial features detection. This system had competitive accuracy (0.95 ± 0.13), runs in real-time (23.23 ± 0.60 fps) [PQA18].

In order to achieve real-time detection performance on embedded systems with limited computing power, this paper used one stage detector YOLOv3 algorithm to achieve face recognition, and some experimental optimizations are made for the YOLOv3 algorithm depend on the task. They optimized the computational efficiency using TensorRT. TensorRT is a C++ library from NVIDIA; it is used for high-performance inference on NVIDIA GPUs and deep learning accelerators. They used an Nvidia GTX 1080 Ti GPU to train the model through the Darknet framework³¹ in an Ubuntu 18.04 environment. During training, They used the SGD with a batch size of 16, the momentum and weight decay are respectively set as 0.95 and 0.005, and adopt batch normalization [THQH22].

This paper proposes a multi-target real-time detection, tracking, and recognition algorithm, which includes three strategies: fast detection, fast-tracking, and quick recognition. First, the human faces are detected by a rapid detection strategy, and MMFNet is used to extract features quickly. Finally, fast face tracking through FTNet. The algorithm avoids the repeated detection and recognition of the same target, and the tracking is stable. It successfully converts the multi-target scene into a single target scene. FTNet achieved high tracking accuracy on OTB and 300VW, and the tracking speed reached 102fps on the Xiph and camera data test set, which improved the tracking speed while ensuring high tracking accuracy. Although MMFNet has reduced efficiency in face recognition, the model size reduced by 45%, and the recognition speed has increased by 25%. The experimental results indicate that the strategy proposed in this paper balances out the problem of time-consuming multi-target tracking and detection in multi-target scenarios. It has sufficient accuracy and real-time performance in detecting and identifying human faces in multi-target scenes [LWFZ20].

In this thesis a system was proposed and implemented to perform real time facial detection and recognition using a drone, in conjunction with a computer, to find and track a specified target individual. The facial recognition systems

were found to be the key component of the entire system, so three different facial recognition systems were implemented to quantify their performance. All though, the different facial recognition systems had varying strengths and weaknesses, FaceNet performed best when the system was tested in a real-world setting. The results of this thesis indicate that using facial recognition algorithms in conjunction with a drone with the goal of tracking an individual is possible. Even when using low cost, light weight drones, facial recognition is possible by having the drone stream video to a computer powerful enough to handle facial recognition. By streaming video to a computer, this system solved many of the performance issues mentioned in papers that implemented similar systems [2;7]. There exist extraneous factors which limit the capabilities of this system; but it served as a strong foundation that can be built upon [MM23].

In this paper, we implement a real-time surveillance consist of face recognition and invader tracking which based on OpenCV. OpenCV provides a set of open source programming libraries which suitable for real-time surveillance system. In this paper, we propose to use AdaBoost Cascade combined with skin model for face detection and then in order to recognize the candidate faces, they will be analyzed by the hybrid Wavelet and LDA method. After that, Mean shift and Kalman filter will be invoked to track the face. The user interface provides the current state, camera view and records for cross-platform user. The experimental results show that the algorithm has quite good performance in terms of real-time and accuracy [LLC⁺10].

Biometric systems are structures that have been often used in recent years. This biometric system is based on the use of some physiological features of the person for security. It is predicted that biometric systems will be an indispensable part of the information security systems in the coming years. Thus, people won't have to carry cards and/or memorize their own passwords. In this study, a biometric system for human face detection and recognition is put into practice. This system put into practice works real time and successfully carries out face recognition, detection and tracking. In this way, the system has the feature of being integrated into different practices. In this study, face detection and tracking system, is put into practice in a personal automation. Here, the aim is to carry out personnel's entry and exit automatically and safely. In this system practiced basing on PCA, in the future studies, various optimization algorithms can be used in order to increase system performance [BD13].

It has been certainly made out from the result and analysis that the proposed work has provided a better implementation of the current algorithms with integration of the modified work. The proposed work of tracking people in live video is just small part of the much larger picture. Compared to legacy systems, assay of facial expressions can help expedite a lot of manual labor that goes into the respective area of work, unnecessarily. Similar approach can be used to further develop system which detect crimes, identify mood of people, security systems, etc. Live Subject tracking, Facial Recognition and Analysis is a much deeper subject which provides immense research opportunities and also development for the greater good of the society [RLJD18].

Chapter 3

FEASIBILITY STUDY

A feasibility study for a Facial Attendance System is crucial to assess the viability of implementing such a system within an organization. The study typically examines various aspects to determine if the proposed solution is feasible from technical, economic, operational, and scheduling perspectives. Here are key components of a feasibility study for a Facial Attendance System:

3.0.1 Technical Feasibility

3.0.1.1 Requirements Analysis

Assess the technical requirements of implementing a Facial Attendance System, including the necessary hardware, software, and networking components.

3.0.1.2 System Compatibility

Evaluate the compatibility of the proposed system with existing infrastructure, databases, and devices.

3.0.1.3 Facial Recognition Technology

Investigate the reliability and accuracy of available facial recognition algorithms and technologies.

3.0.2 Economic Feasibility

3.0.2.1 Cost-Benefit Analysis

Evaluate the costs associated with developing, implementing, and maintaining the Facial Attendance System. Compare these costs with the expected benefits, including time savings, accuracy improvements, and potential cost reductions.

3.0.2.2 Return on Investment (ROI)

Calculate the expected ROI over a specified period, considering both tangible and intangible benefits.

3.0.3 Operational Feasibility

3.0.3.1 User Acceptance

Assess the willingness and ability of users (both administrators and employees) to adapt to and accept the new system.

3.0.3.2 Training Requirements

Identify the training needs for users and administrators to effectively use and manage the Facial Attendance System.

3.0.3.3 Integration with Existing Processes

Evaluate how seamlessly the system can integrate with existing attendance tracking and HR processes.

Chapter 4

Methodology

4.1 Software Development Methodology

An iterative and incremental software development approach, such as Agile methodology, can be adopted. Agile methodologies emphasize flexibility, collaboration, and iterative development, which align well with the nature of the project and its potential changes and refinements.

4.1.1 Agile Methodology

Agile development is a flexible and iterative approach to software development that prioritizes adaptability and collaboration. It emphasizes delivering small, incremental releases of a product with the goal of responding quickly to changing requirements and feedback. Agile methodologies are based on the Agile Manifesto, which values individuals and interactions, working solutions, customer collaboration, and responding to change over rigid processes and tools.

The Agile development process is typically organized into iterations or sprints, each of which includes the following key steps:

- **Requirements:** In Agile, requirements are expressed as user stories, which are short descriptions of a feature told from the perspective of the end user. Typically, the product owner collaborates with stakeholders to define and prioritize user stories, which in our case, are the objectives and highlights of the problem statement. These stories are added to the product backlog.
- **Design:** Design involves planning how the requirements will be implemented, considering the overall architecture, user interface, and other relevant aspects. During sprint planning, the development team discusses and plans how to address the problem statements and achieve the objectives. Design decisions are made collaboratively, and the team may create mock-ups or prototypes to visualize the solution.
- **Development:** This phase involves coding or implementing the solutions outlined in the design phase. Development work is carried out based on the

tasks assigned to team members during the sprint planning. Continuous collaboration and communication are crucial to address any issues or changes that may arise.

- **Testing:** Testing is an integral part of Agile development and involves validating that the implemented features meet the specified requirements. Testing will be done parallelly with development, creating test cases based on user stories and conducting various testing types (unit testing, integration testing, etc.) to ensure the quality of the code and output.
- **Deploy:** Deployment involves making the developed features available for use by end-users. The product increment is deployed to a staging or production environment. Continuous integration and deployment practices ensure that the latest changes are automatically integrated and deployed when ready.
- **Review:** At the end of each iteration, a review or demo is conducted to showcase the completed features to stakeholders. The team will participate in a review meeting to provide feedback, discuss any adjustments needed, and reprioritize the product backlog.
- **Launch:** The final step involves releasing the product increment to users. In our project, a web interface will be rolled at launch which will facilitate user interaction. Once all necessary adjustments are made based on feedback and the priorities based on objectives the product is officially launched. The cycle then repeats with a new iteration.

It's important to note that the design, development, testing, deployment, review, and launch steps occur iteratively and multiple times throughout the project. Agile development promotes continuous improvement and adaptation to changing requirements, allowing teams to deliver value to users more frequently and efficiently.

4.2 ALGORITHMS AND MODELS

Facial recognition systems rely on various algorithms and models to accurately identify and verify individuals based on facial features. Here are some common algorithms and models used in facial recognition systems:

4.2.1 Haar Cascades

1. **Description:** Haar Cascades is an object detection algorithm used for identifying faces in images or video streams. It works by training on positive and negative images to create a classifier.

2. Application: Initial face detection before more detailed analysis.

4.2.2 Local Binary Patterns

- Description: LBP is a texture descriptor that is often used for facial feature extraction. It focuses on local patterns of pixel intensities.
- Application: Feature extraction in facial recognition systems.

4.2.3 Eigenfaces

1. Description: Eigenfaces is a facial recognition algorithm that uses Principal Component Analysis to represent facial features in a lower-dimensional space
2. Application: Recognizing faces based on principal components.

4.2.4 Local Feature Analysis

1. Description: LFA focuses on extracting local features, considering facial regions independently. It enhances robustness against variations.
2. Application: Effective in handling facial variations due to expressions, pose, and lighting.

4.2.5 Convolutional Neural Networks

1. Description: CNNs are deep learning models that automatically learn hierarchical features from raw pixel data. They excel at capturing spatial hierarchies in images.
2. Application: State-of-the-art facial recognition, especially when dealing with large datasets and complex variations.

4.3 Proposed System

This system will be able to detect student or employee without physical contact and update the database based on the response from system. The system architecture of the proposed system is given below,

Figure 4.1: System architecture of face recognition system

4.3.1 Dataset Creation and Image Preprocessing

Images of Employee or student are captured using a web cam. Multiple images of single student will be acquired with varied gestures and angles. These images undergo pre-processing. The images are cropped to obtain the Region of Interest (ROI) which will be further used in recognition process. Then these images will be converted from RGB to gray scale images. This is done using OpenCV.

1. Step1: we need to read the image with OpenCV's `imread()` function:

```
img = cv2.imread(imagePath)
```

This will load the image from the specified file path and return it in the form of a Numpy array.

```
img.shape
```

Output:

```
(1000,500,4)
```

2. Step2: Convert the RGB data of Image into Greyscale To improve computational efficiency, we first need to convert this image to grayscale before performing face detection on it: Code

```
gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2_GRAY)
```

Output:

```
(1000,500)
```

This array only has two values since the image is grayscale and no longer has the third color channel.

4.3.2 Face Detection

The idea behind this technique involves using a cascade of classifiers to detect different features in an image. Haar Cascade algorithm needs to be trained to detect human faces before it can be used for face detection. This is called feature extraction. The haar cascade training data used is an xml file- `haarcascade_frontalface_default`. We are using a file called `haarcascade_frontalface_default.xml`. This classifier is designed specifically for detecting frontal faces in visual input. `face_classifier = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")`. We are using `detectMultiScale` module from OpenCV. This is required to create a rectangle around the faces in an image. It has got three parameters to consider- `scaleFactor`, `minNeighbors`, `minSize`.

4.3.3 Face Recognition and Calculation

Face recognizer used in this system is Local Binary Pattern Histogram. Initially, the list of local binary patterns (LBP) of entire face is obtained. These LBPs are converted into decimal number and then histograms of all those decimal values are made. At the end, one histogram will be formed for each images in the training data. Later, during recognition process histogram of the face to be recognized is calculated and then compared with the already computed histograms and returns the best matched label associated .

4.3.4 DataBase

After face recognition process, Based on Calculated histogram Databases are updated. The name of employee or student along with day and time of attendance is also be stored in the database.

4.4 System (or Project) Design and Architecture

We developed a Facial Recognition Attendance System which takes Face image as its input and process it to provide the value of matching face as its output. Then, it take decision of present and absent person.

4.5 USE CASE DIAGRAM

Figure 4.2: System Use Case Diagram

4.6 System Block

Figure 4.3: System Block Diagram

4.7 Sequence Diagram

Figure 4.4: Sequence Diagram

4.8 Class Diagram

Figure 4.5: Class Diagram

Chapter 5

Implementation Plan

5.1 Schedule(Gantt-Chart)

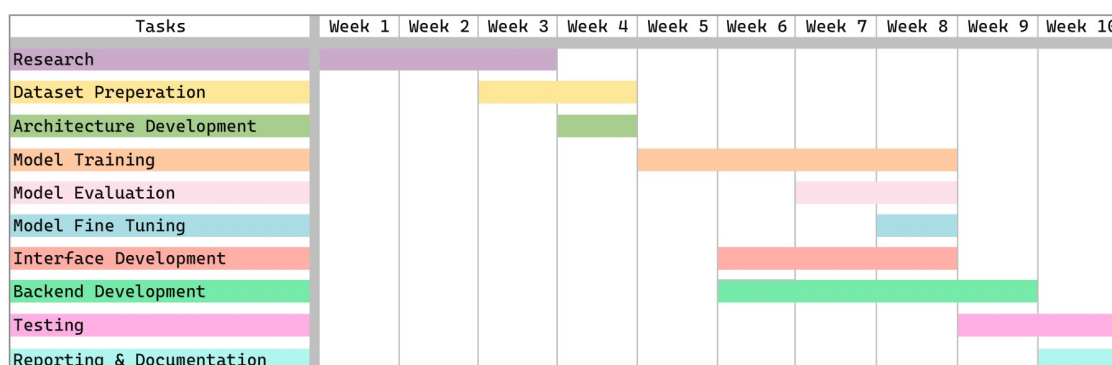


Figure 5.1: Gantt Chart

5.2 SOFTWARE REQUIREMENT

Our Facial Regonition system requires Python, OpenCV, MySQL, TensorFlow, Flask, CNN, JavaScript, Keras which are described below;

5.2.1 Python

Python is a high-level interpreted programming language for general-purpose programming created by Guido van Rossum which as released in 1991. Python has design philosophy that emphasizes code readability, notably using significant white spaces. It provides constructs that enable clear programming in small and large scales. The programming language features dynamic type system and automatic memory management with support to multiple programming paradigms including Object Oriented Imperative, Functional and Procedural. The programming language has large comprehensive standard library.

5.2.2 OpenCV (Open Source Computer Vision Library)

OpenCV is a popular open-source library designed for computer vision tasks. It provides a wide range of functions and algorithms related to image and video processing. OpenCV is widely used for tasks like image and video processing, object detection, facial recognition, and more. It is written in C++ but has Python bindings, making it accessible to a broader audience.

5.2.3 MySQL

MySQL is an open-source relational database management system (RDBMS). Its name is a combination of "My", the name of co-founder Michael Widenius's daughter, and "SQL", the abbreviation for Structured Query Language. In our project, we use this database to feed into PHP system for data labeling. The database is created from scraping Nepali Sentiment text in python.

5.2.4 Tensor Flow

TensorFlow is an open-source deep learning framework developed by the Google Brain team. It is widely used for building and training machine learning and deep learning models. It provides Flexible architecture for building various machine learning models. It supports deep learning, including neural networks and convolutional neural networks (CNNs). TensorFlow is used for developing and training deep learning models for tasks such as image classification, object detection, natural language processing, and more.

5.3 FUNCTIONAL REQUIREMENT

5.3.1 User Registration

The system should allow administrators to register individuals by capturing and storing their facial features. Registration should be straightforward and user-friendly.

5.3.2 Face Detection

Accurate face detection algorithms should be implemented to identify and locate faces in images or video streams.

5.3.3 Face Recognition

Employ robust facial recognition algorithms for accurate identification of registered individuals. The system should provide a confidence score or percentage to indicate the level of certainty in the recognition process.

5.3.4 Enrollment and Database Management

The system should manage a secure database of enrolled users and their corresponding facial features. Allow easy addition, modification, or removal of users from the database.

5.3.5 Real-time Monitoring

Provide real-time monitoring capabilities for administrators to view attendance status. Generate alerts for late entries or other specified conditions.

5.4 NON-FUNCTIONAL REQUIREMENT

These requirements are not needed by the system but are essential for the better performance of sentiment engine. The points below focus on the non-functional requirement of the system.

5.4.1 Performance

5.4.1.1 Response Time

The system should have low latency in recognizing faces and recording attendance.

5.4.1.2 Throughput

It should be able to handle a specified number of facial recognition transactions per second.

5.4.2 Scalability

The system should be scalable to accommodate an increasing number of users and devices.

5.4.3 Reliability

The system should be highly reliable, minimizing the risk of false positives or negatives in facial recognition. It should have mechanisms to handle system failures and recover gracefully.

5.4.4 Availability

The system should have a high level of availability to ensure that it is accessible when needed.

5.4.5 Security

5.4.5.1 Data Security

Facial data and attendance records should be stored and transmitted securely, following industry best practices.

5.4.5.2 Access Control

Implement robust access control mechanisms to restrict unauthorized access to the system.

Chapter 6

Expected Outcomes

The users can interact with the system using a GUI. This method will have higher accuracy in recognition of multiple faces from a single frame with lower response time.

S.N.	Model	Testing Accuracy
1.	Support Vector Machines (SVMs)	87%
2.	Multilayer Perception (MLP)	86.5%
3.	Convolutional Neural Network(CNN)	98%

The best model was found to be with Convolutional Neural Network(CNN) testing accuracy 98%. Other models performed worse than this model is due to,CNN functions automatically as a feature extractor, whereas ML algorithms function as a classifier After building the basic CNN model we tried to increase its accuracy by combining 2D and 3D face recognition techniques, or using different machine learning algorithms to recognize faces more accurately. Other problem is to detect high variability of human faces in terms of shape, size, pose, expression, illumination, occlusion, and makeup. Haar Classifier technique can be adapted to accurately detect facial features. However, the area of the image being analyzed for a facial feature needs to be regionalized.

6.1 Conclusion

The implementation of a Facial Recognition Attendance System using OpenCV aligns with technological advancements, offering a more accurate and efficient solution for attendance tracking. This proposal lays the foundation for a robust system that addresses the current challenges in attendance management.

Bibliography

- [BD13] Muhammet Baykara and Resul Das. Real time face recognition and tracking system. pages 159–163, 11 2013.
- [LCSH22] Teh-Lu Liao, Hsin-Chieh Chen, Qing-Huang Song, and Yi-You Hou. Face recognition and real-time tracking system based on convolutional neural network and parallel-cascade pid controller. *Measurement and Control*, 55(7–8):616–630, July 2022.
- [LLC⁺10] Shou-Jen Lin, Chao-Yang Lee, Mei-Hsuan Chao, Chi-Sen Chiou, and Chu-Sing Yang. The study and implementation of real-time face recognition and tracking system. In *2010 International Conference on Machine Learning and Cybernetics*, volume 6, pages 3050–3055, 2010.
- [LWFZ20] Jun Li, Yaoru Wang, Guokang Fang, and Zhihui Zeng. Real-time detection tracking and recognition algorithm based on multi-target faces. *Multimedia Tools and Applications*, 80(11):17223–17238, August 2020.
- [MM23] Artak Melkumyan and Katya Mkrtchyan. Real time facial recognition and tracking system using drones. In *2023 IEEE 20th Consumer Communications Networking Conference (CCNC)*, pages 975–976, 2023.
- [PQA18] Wesley L. Passos, Igor M. Quintanilha, and Gabriel M. Araujo. Real-time deep-learning-based system for facial recognition. *Anais De XXXVI Simpósio Brasileiro De Telecomunicações E Processamento De Sinais*, January 2018.
- [RLJD18] K. Bharath S Reddy, Onkar Loke, Shantanu Jani, and Kanchan Dabre. Tracking people in real time video footage using facial recognition. *2018 International Conference on Smart City and Emerging Technology (ICSCET)*, January 2018.